A Patched-Conic Mission Analysis Tool

Lunar & interplanetary design — approach, validation, limitations

MECE 4305 Mechanics & Thermodynamics of Propulsion — Week 10

9 November 2026

Columbia University, Fall 2026 • `mission_analysis` (`sfmission`) v0.1

Why: the first four steps of the design roadmap

1. **Mission definition** — target, dates, orbits
2. **Required Δv** — launch, injection, flybys, capture, landing
3. **Propellant mass fraction $\rightarrow$ required I_{sp}**
4. **Staging / architecture**

5–10. thrust, thrust chamber, $\dot{m}$, V_e, p_c, M_e, A_e, feed system, iterate (Weeks 11–12)

The tool computes steps 1–4 and hands a `DeltaVBudget` with propellant masses and required I_{sp} to the engine-design steps.

Three uses

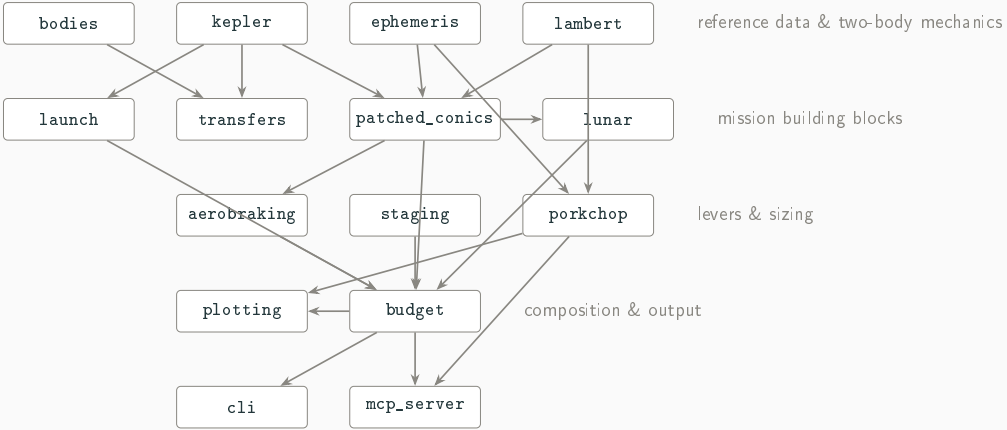
- Week-10 lecture instrument (live porkchop of the window opening *this month*)
- HW 1 reference solution and the first module of each team's project model
- Preliminary design tool exposed to agents via MCP

What it computes

Launch	Δv to orbit: $\lvert v_{orbit} - v_{site} \rvert$ with azimuth for inclination, loss band
Transfers	Hohmann, bi-elliptic, plane change (optimal split), phasing, round trips
Ephemeris	JPL mean elements (1800–2050; 3000 BC–3000 AD), truncated-Meeus Moon
Lambert	universal-variable, bracketed root; type I/II, prograde/retrograde
Patched conics	departure & capture burns, unpowered & powered flybys, multi-leg sequences
Porkchops	C_3, v_∞^{arr}, Δv grids; optima; window spans; multi-year scans
Earth–Moon	Hohmann estimate; Bate–Mueller–White patched conic with perilune design solver
Aero levers	aerocapture credit; per-pass aerobraking Δv; campaigns; integrated pass check
Staging	rocket equation, required I_{sp}, closed-form optimal N-stage split
Budgets	legs with bands and margins $\rightarrow$ propellant mass, ζ, required I_{sp}
Front ends	Python API, CLI (`sfmission`), 23 MCP tools (`sfmission-mcp`)

Fidelity class: patched conics, impulsive burns, analytic ephemerides — sizing accuracy, not navigation accuracy.

Architecture



Pure functions returning dataclasses; km, s, rad internally; degrees for `*_deg` arguments; epochs as ISO strings or JD. 85 tests run in 1 s.

Approach I — reference data and their accuracy

- *GM*: JPL DE440 (planet-*system* values for outer planets); IAU radii; Laplace SOI $a(m/M)^{2/5}$
- Planets: JPL *Keplerian Elements for Approximate Positions* — six elements + secular rates, J2000 ecliptic
- Moon: truncated Meeus Ch. 47 (32 lon/dist, 20 lat terms), precessed to J2000
- "Earth" = Earth–Moon barycenter unless corrected

vs. astropy	angle [″]	rel. dist.	v [m/s]
Inner planets	< 35	$< 1.5 \times 10^{-4}$	< 5
Jupiter	< 340	$< 4 \times 10^{-4}$	< 28
Saturn	< 560	$< 1.5 \times 10^{-3}$	< 41
Uranus/Neptune	< 110	$< 1 \times 10^{-3}$	< 16
Moon	< 40	< 40 km	–

Consequence for design: v_∞ errors ~ 1 m/s at Earth, ~ 10 m/s at Saturn — far below budget margins.

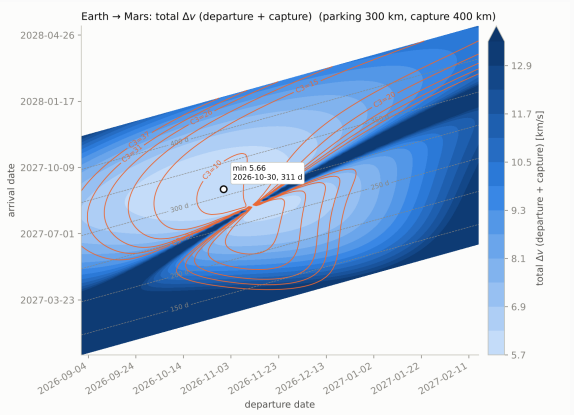
Approach II — Lambert arcs and porkchop maps

Universal-variable Lambert (BMW/Curtis):

$$F(z) = \left[\frac{y}{C}\right]^{3/2} S + A\sqrt{y} - \sqrt{\mu}\,\Delta t = 0$$

- F monotonic on the single-rev branch $\Rightarrow$ bracket + Brent: never diverges
- type I ($< 180°$) and type II fall out of the prograde geometry
- 55 µs per solve; 110^2 grid in $\sim$1 s
- each cell: C_3, v_∞^{arr}, Δv_{dep}, Δv_{cap}, DLA

Earth→Mars 2026/27 optimum: depart 2026-10-30, 311 d, $C_3 = 9.2$, $\Delta v = 5.66$ km/s (300 km parking → 400 km Mars orbit).



Approach III — patched conics

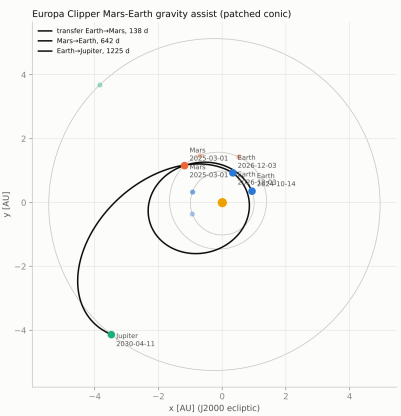
Departure / capture (tangential periapsis burns)

$$\Delta v_{dep} = \sqrt{v_\infty^2 + \tfrac{2\mu}{r_p}} - \sqrt{\tfrac{\mu}{r_p}}, \quad \Delta v_{cap} = \sqrt{v_\infty^2 + \tfrac{2\mu}{r_p}} - v_p^{orbit}$$

Unpowered flyby

$$e = 1 + \frac{r_p v_\infty^2}{\mu}, \quad \delta = 2\arcsin\frac{1}{e}, \quad |\Delta v_{helio}| = 2v_\infty \sin\frac{\delta}{2}$$

Powered flyby (MGA model): shared r_p with $\frac{1}{2}\delta_{in} + \frac{1}{2}\delta_{out} = \angle(\mathrm{v}_\infty^{in}, \mathrm{v}_\infty^{out})$, speed mismatch burned at periapsis; infeasible if $r_p < r_{min}$.

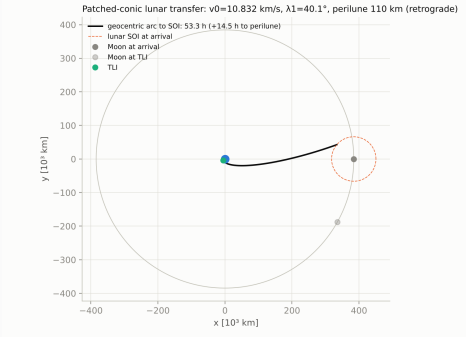


Europa Clipper dates: C_3 41 vs. 77 direct; injection 4.95 vs. 6.30 km/s; powered corrections 0.13 + 0.13 km/s at Mars and Earth.

Approach IV — Earth–Moon patched conic (Bate–Mueller–White §7.4)

- Geocentric ellipse from (r_0, v_0, ϕ_0) to the SOI point set by λ_1
- $v_2 = v_1 - v_{Moon} \rightarrow$ selenocentric hyperbola (e, r_p, v_p)
- **Vector implementation**: sense of motion from $\mathrm{sign}(h_2)$, no trig sign ambiguities (scalar relations reproduced in tests)
- Adds the **SOI transit time** (14–18 h) — essential to compare with flown data
- Design solver: λ_1 for given v_0 (or v_0 for λ_1) hitting a perilune altitude; prograde and retrograde roots

Hohmann estimate: TLI 3.135, LOI 0.82 km/s; 3-day transfer: TLI 3.157, LOI 0.889 km/s.



Approach V — launch Δv and the aero levers

Launch

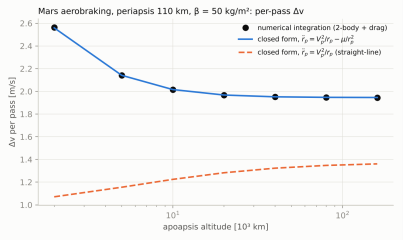
$$\Delta v = |\mathrm{v}_{orbit} - \mathrm{v}_{site}| + \Delta v_{grav} + \Delta v_{drag} + \Delta v_{steer}$$

with $\mathrm{v}_{site} = \omega R \cos\varphi\, \hat{\mathrm{e}}_E$ and $\cos i = \cos\varphi \sin Az$. KSC, 300 km, due east:
$7.726 - 0.409 + (1.15 \ldots 2.05) = 8.47\,\mathrm{km/s}$ to $9.37\,\mathrm{km/s}$.

Aerobraking, per pass (exponential atmosphere, parabolic $r(t)$ about periapsis)

$$\Delta v_{pass} = \frac{\rho_p V_p^2}{2\beta}\sqrt{\frac{2\pi H}{\ddot{r}_p}}, \qquad \ddot{r}_p = \frac{V_p^2}{r_p} - \frac{\mu}{r_p^2}$$

Dropping μ/r_p^2 (straight-line fly-through) gives the Week-11 slide form, low by $\sqrt{V_p^2/(V_p^2 - \mu/r_p)} \to \sqrt{2}$; integration confirms the gravity-aware form to 0.2%.



Aerocapture credit at Mars ($v_\infty = 2.65$):
2.08 km/s replaced by a 0.06 km/s periapsis raise.

Approach VI — rocket equation, staging, budgets

Sutton nomenclature: $\zeta = m_p/m_0$, $MR = m_f/m_0$, $\sigma_k = m_{s,k}/(m_{s,k} + m_{p,k})$, $c_k = g_0 I_{sp,k}$, $r_k = e^{\Delta v_k / c_k}$ (ε reserved for area ratio)

$$\frac{m_0}{m_L} = \prod_k \frac{r_k(1-\sigma_k)}{1 - r_k\sigma_k}, \qquad I_{sp}^{req} = \frac{\Delta v}{g_0 \ln[1/(1-\zeta)]}$$

Optimal split for fixed $\sum \Delta v_k$ (one scalar root in λ):

$$r_k = \frac{\lambda c_k - 1}{\lambda c_k \sigma_k}, \qquad \sum_k c_k \ln r_k = \Delta v_{tot}$$

`DeltaVBudget`

ordered legs with nominal/low/high, kind (propulsive / aero / margin), optional stage I_{sp}

- composers: dated interplanetary (launch, injection, flybys, capture or aerocapture, landing, margin); lunar (TLI, LOI, descent, ascent, TEI)
- backward propellant sizing; ζ and required I_{sp} — the hand-off to engine design

Validation — textbook, independent models, flown missions

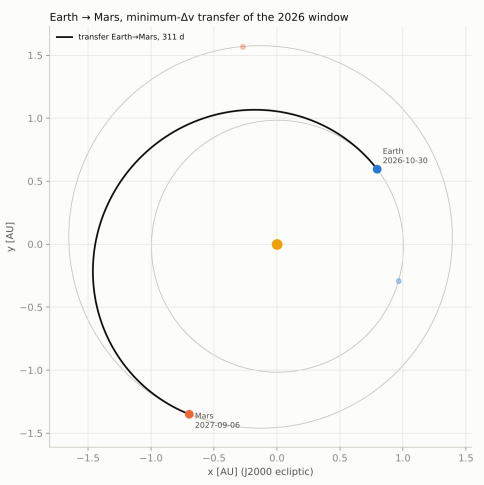
Lambert, Curtis Ex. 5.2; 20 closure geometries	v_1, v_2 to 1×10^{-4} km/s; closes to 1×10^{-8} km
Earth→Mars Hohmann (Curtis Ex. 8.4–8.6)	v_∞ 2.943/2.648, Δv 3.590 km/s, phase 44.3°, synodic 779.9 d
LEO→GEO; +28.5° plane change	3.893 km/s; 4.23 km/s
Mars 2020 actual dates	path 470.9 Mkm vs. ≈471 Mkm (JPL); C_3 14.5; v_∞^{arr} 2.55 km/s
Voyager 2 E–J–S dates	Jupiter turn 97°; r_p 620,000 vs. ≈570,000 km; powered fix 0.05 km/s; C_3 105
Apollo 11 published TLI state	TLI→perilune 68–73 h vs. 73.0 h; LOI 0.89–0.92 vs. 0.891 km/s
Aerobraking pass vs. integration	1.952 vs. 1.951 m/s
Ephemeris vs. astropy; Mars oppositions	table on slide 5; longitudes agree $< 1.5°$
Staging identities, optimal split	exact

Lecture point from Apollo 11: the two published TLI rows differ by $0.04\,\mathrm{km}^2/\mathrm{s}^2$ in energy, and lunar TOF moves ∼5 h per $0.1\,\mathrm{km}^2/\mathrm{s}^2$ — the result is as good as the altitude datum.

Utility — numbers for Week 10

Lever (Mars v_∞ = 2.65 km/s)	without	with
Aerocapture, 400 km orbit	2.08 km/s	0.06 km/s
Aerobraking 110×40,000→400 km	2.08 km/s	0.92 km/s, 152 d
Mars–Earth assist to Jupiter	C_3 77, 2.7 yr	C_3 41, 5.5 yr
KSC credit, east vs. polar	0.41 km/s	−0.01 km/s

- Earth–Mars phase angle on 11/12: 56° vs. 44.3° Hohmann requirement
- Type II (311 d, 5.66 km/s) vs. type I (260 d, 6.02 km/s) in this window
- Next windows: 2028-11, 2030-12/2031-02, 2033-03/04 (type II / type I pairs)

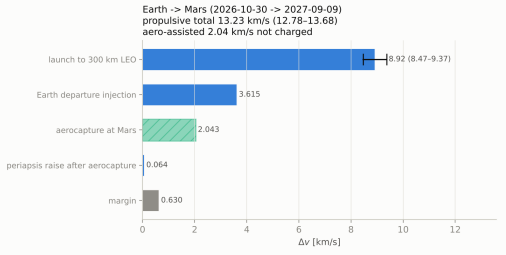


Utility — HW 1 reference chain (steps 1–4 end to end)

Mars orbiter, 2026 window, KSC:

1. launch band 8.47–9.37 km/s
2. dated legs at the porkchop optimum
3. budgets: post-launch Δv **6.39** (propulsive) vs. **4.31** km/s (aerocapture)
4. required I_{sp} at $\zeta = 0.7$: 541 s vs. 365 s
5. 1000 kg final mass, 320-s engine $\rightarrow$ 7,653 kg injected
6. two-stage launcher (I_{sp} 300/450, σ 0.07/0.10) for 12.5 km/s: gross 695 t, optimal split 4.8/7.7 km/s

Teams swap mission, dates and engines; the `DeltaVBudget` feeds the Week 11–12 thermochemistry and engine modules.



Limitations I — physics

- **Patched conics**: no third-body gravity on a conic, sharp SOI. Few-% error in planetocentric burns for planetary missions; hours of TOF and tens of m/s of LOI for the Moon (Apollo 11: 0–7%).
- **Impulsive burns**: finite-burn losses of low-T/W stages not charged; launch losses are a user band, not a trajectory.
- **Ephemeris**: arcsec–arcmin mean elements; Earth = EM barycenter by default. Fine for Δv, not for targeting.
- **Circular assumptions**: Moon orbit circular at 384,400 km (perigee–apogee $\pm$21,000 km $\Rightarrow \pm$10 m/s TLI, hours of TOF); quick estimates use circular coplanar planets.
- **Plane/DLA**: 3-D legs, but no plane-change penalty for high-DLA departures or parking-orbit incompatibility; DLA reported only.
- **Lambert**: single revolution; 180° transfers undefined (NaN ridge).
- **Powered flyby**: shared-periapsis tangential-burn model is a lower bound; flyby dates are inputs, not optimized.

Limitations II — scope

Aero-assist

- aerocapture charged as a Δv credit only: no corridor, heating, g-load, TPS mass, guidance
- single-scale-height atmosphere fitted to 90–150 km; constant β; Mars density varies $\times 2 \Rightarrow$ pass count and duration do too

Staging

- σ is an input; no T/W, tank or residual models; σ_k independent of stage size

Reference values, not computed

- lunar descent/ascent 2.07/1.86 km/s (Apollo LM); Mars landing Δv user-supplied

Not covered

- low-thrust / electric trajectories
- launch-vehicle performance curves (C_3 vs. payload)
- navigation and statistical Δv, environments, operations

Next steps and how to run it

Planned (priority order)

1. low-thrust legs: Edelbaum Δv, power-limited sizing (HW 3b, EP teams)
2. DE440 option + perturbed propagator to quantify patched-conic error on request
3. multi-revolution Lambert
4. launch-vehicle performance curves
5. coupling to the ascent-trajectory optimizer (computed launch band)

```
pip install -e ".[dev,mcp]"      # repo root
pytest tests/mission_analysis    # 85 tests

from modern_aerospace_propulsion.mission_analysis
     import *
pc = porkchop("Earth","Mars
     ","2026-09-01","2027-02-15",120,450)
print(pc.summary())
b, m = interplanetary_budget("Earth","Mars",
        "2026-10-30","2027-09-06", aerocapture=True
     )
print(b.table()); b.size_propellant(1000, 320)

sfmission lunar --tof 72 --plot lunar.png
sfmission-mcp     # {"mcpServers":{"sfmission":
                  #   {"command":"sfmission-mcp"}}}
```

Code, tests, examples, figures, report and these slides: `Week 10 - Mission Analysis/` in the course repository.